Daniel Liu

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About Me

Professional Interests: High Performance Computing, Distributed Systems, Quantitative & Decentralized Finance
Languages: Python, C++, C (GCC/clang), Matlab, TypeScript
Technologies: Git, Unix, Vi, screen, regex/grep, Jupyter, XGBoost, SQL, Elasticsearch, Redis, Flask, Celery, Apache Airflow
Legal Status: Canadian Citizen. Available on TN/H1B visa. Green Card in progress.

Experience

Terrestrial Energy
Toronto
Software Engineer, Computational Physics
January 2020 – Present

- Owner of developing next-generation reactor core physics parameters using monte carlo physics algorithms in Python & C
- Migrating legacy simulation programs to compile CPU/GPU multi-threading support by adding code directives per OpenMP C/C++ specifications
- Optimizing heavily CPU-dependent physics programs to ensure data structures are vectorization-friendly to exploit SIMD processing. Improved C/C++ code runtimes to decrease CPU hours by 10-50% in between simulations.
- Testing expert for the team, writing guides, and giving presentations on new internal testing tools and best practices

Ontario Power Generation
Toronto
Software Engineer, Reactor Simulations
September 2018 – December 2019

- Built a notification worker portal that organizes and manages decontamination campaigns for all nuclear reactors cores in Canada using HTML, CSS, JS, Airflow, Google Cloud Services & physics models
- Created a real time Python service that tracks and publishes logic decisions for plant control
- Improved large dataset performance and core features for preventative maintenance using WebSockets, C++ and JavaScript
- Optimized SQL database queries on multiple ODBC connections to push maintenance notices on critical nuclear equipment

Canadian Nuclear Laboratories
Toronto
Software Engineer, Nuclear Simulations
January 2018 – September 2018

- Designed code to model the flow of groundwater surrounding below-grade nuclear waste repositories
- Implemented additional modules describing the interaction of dissolved chemical species with the surrounding earth
- Optimized aggregation and filter queries performed on virtual tables by strategically rejecting rows before computing expensive requests, using generators to reduce process memory usage by ∼80%
- Created MATLAB code to simulate gas-catalyst interactions to develop and scale-up an experimental process to separate Tritium from other gasses, a radioactive isotope of Hydrogen used for fusion energy research.

Personal Projects

10-Q/10-K Insights : Built on-top of a scraper to filter all 10-Q/K earning reports to predict future returns by algorithmic scoring of fraudulent statements from SEC EDGAR. Deployed algorithm with an AUM of $3 M USD. Used python, pandas, XBRL

McKinsey & Company Hackathon : Analyzed if a requested loan is approved or not, based on the features of the applicant (e.g., income, gender, city), and that of the loan (e.g., loan amounts, interest rates). Used XGBoost / gradient boosting for analysis

Scotiabank Hackathon : Concepted an addicting mobile game to help reduce user debt with in-app purchases routed back to saving accounts. Participated with seven software engineers and graphics designers within 48 hours to win a $10,000 prize

Education

University of Ontario Institute of Technology *2018*
Bachelor of Engineering – Nuclear Engineering, Minors in Computer Science (95% cGPA)

Research : Created a MATLAB application to track AR device user location in GPS denied environments using MEMS input sensor data on Google Glass.